ETHERS

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Ethers

Ethers :

The simplest ether is dimethylether and the commonest ether is diethylether, also called sulphuric ether.

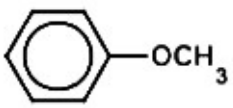
In the IUPAC system, ethers are named as alkoxyalkanes. The smaller alkyl group along with –O– is named as alkoxy group and the name of the parent hydrocarbon of the other alkyl group succeeds it. Position of alkoxy group in the parent is indicated.

Introduction

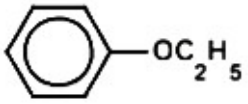
Ethers are alkyl or aryl derivatives of alcohols or phenols. They are formed by the substitution of –H in –OH group of alcohol or phenol. Ethers are also treated as dialkyl derivatives of water. The general formula of ethers is R–O–R', where R and R' are alkyl, alkenyl or aryl groups.

Aromatic ethers:

Methyl phenyl ether is called anisole. Its IUPAC name is methoxy benzene.



(Methoxy benzene)

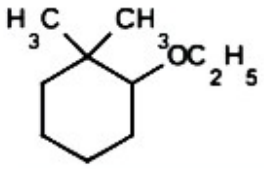


(Methoxy benzene)

Unsaturated ethers :

Allyl methyl ether is 3-methoxypropane. $CH_2=CH-CH_2OCH_3$

Cyclic ethers : Ethylene oxide, also called epoxide is a cyclic ether,

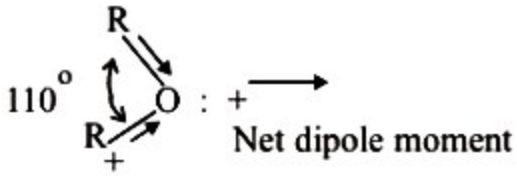
Compound	Common name	IUPAC name
CH_3OCH_3	Dimethyl ether	Methoxymethane
$C_2H_5OC_2H_5$	Diethyl ether	Ethoxyethane
$CH_3OCH_2CH_2CH_3$	Methyl n-propyl ether	1-Methoxypropane
$CH_3OCH(CH_3)_2$	Methyl isopropyl ether	2-Methoxypropane
$C_6H_5OCH_3$	Methyl phenyl ether (anisole)	Methoxybenzene (anisole)
$C_6H_5OCH_2CH_3$	Ethyl phenyl ether (Phenetole)	Ethoxybenzene
$C_6H_5O(CH_2)_6-CH_3$	n-Heptylphenyl ether	1-Phenoxyheptane
$C_6H_5-O-CH_2-CH_2-CH(CH_3)_2$	Phenyl isopentyl ether	3-Methylbutoxybenzene
$CH_3-O-CH_2-CH_2-OCH_3$	Dimethoxy Ethane	1,2-Dimethoxyethane
$CH_3-O-CH=CH_2$	Methyl vinyl ether	Methoxyethene
	---	2-Ethoxy-1, 1-dimethylcyclohexane

General Properties of ethers

(a) The lower members are gases or volatile liquids, and their vapours are highly inflammable.

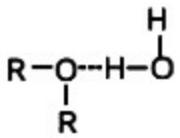
(b) Boiling points of ethers are much lower than those of the alcohols containing the same number of carbon atoms, this is due to the fact that ethers can not associate trough hydrogen bonding.

(c) Since the C–O–C bond angle is not 180°; the dipole moments of the two C–O bonds do not cancel each other, consequently, ethers possess a small net dipole moment (eg 1.18D for diethyl ether).



(d) This weak polarity does not appreciably affect the boiling points of ethers, which are about the same as those of alkanes having comparable molecular weights, and much lower than those of isomeric alcohols. Compare, for example the boiling points of n-heptane (98°C) methyl n-pentyl ether (110°C) and n-hexyl alcohol (157°C).

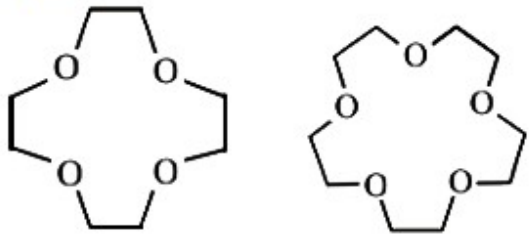
(e) Ethers show a solubility in water comparable to that of the alcohols, both diethyl ether and n-buthyl alcohol for example, being soluble to the extent of about 8 gm per 100 gm of water. The solubility of lower alcohols to water is due to hydrogen bonding between water molecules and alcohol molecules; presumably the water solubility of ether arises in the same way.



As the molecular mass increases, the solubility of ethers in water decreases due to a corresponding increase in the hydrophobicity of the hydrocarbon portion of the molecule. Ethers are however fairly soluble in common organic solvents such as alcohol, benzene, chloroform etc.

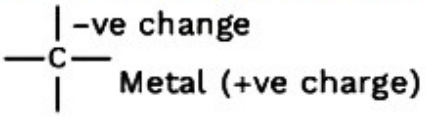
(f) All ethers are lighter than water.

Crown Ethers:
For example



12-crown-4 15-crown-5

Know about Organometallic Compound:

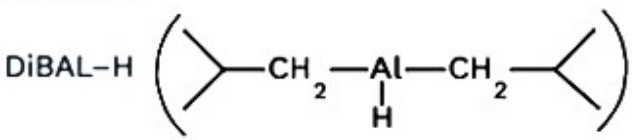


Where metals = Na, Li, Mg, Al, Zn, Cd, Hg etc.

The ionic character order is:
Na > Li > Mg > Al > Zn > Cd > Hg

The reactivity of the organometallic compound increases with the ionic compounds like organosodium (RNa) and organopotassium (RK) are very reactive.
They are really inflammable in air, react violently with water are non-volatile and do not react in non-polar solvents. However, more covalent compounds like tetraethyl lead $(C_2H_5)_4Pb$ and diethyl mercury $(C_2H_5)_2Hg$ are for less reactive and relatively stable in air, volatile and soluble in non-polar solvents.

Example



Ethylmagnesiumiodide $(CH_3–CH_2–Mg–I)$
Ethyl Lithium $(CH_3–CH_2–Li)$
Tetraethyl Lead $(CH_3–CH_2)_4Pb$

Introduction

Crown ethers are large ring compounds containing several oxygen atoms, usually in a regular pattern. These may be regarded as condensation polymers of ethylene glycol. The term crown was suggested by the three-dimensional shape of these molecules. These are named as x-crown-y where x is the total no. of atoms in the ring an y is the number of oxygen atoms.

Introduction

Organometallic Compound are those organic compounds in which a metal is directly joined to carbon. The ionic character of the carbon-metal bond depends on the nature of the metal.

Points to remember

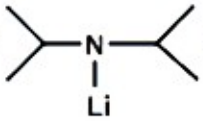
- The extent of ionic character of C–M bond depends on the electropositive nature of the metal.
- "More electropositive the metal, more ionic the C–M bond
- The organometallic compound of Li and Mg, which are of intermediate ionic character are easy to prepare and are used as synthetic reagents.

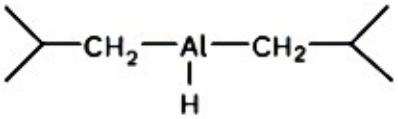
Q1 **Identify the organometallic compounds:**

(A) CH_3CH_2ONa

(B) $NaNH_2$

(C) $H-C\equiv \overset{\ominus}{C}N\overset{\oplus}{a}$

(D) LDA  **(Lithium di-isopropyl amide)**

(E) DiBAL-H 

(F) CH_3MgBr

A1 **(E, F)**

Q2 **Identify the reactivity order:**

(A) CH_3K **(B)** CH_3Na **(C)** CH_3Li **(D)** CH_3MgBr

(E) CH_3CuCl **(F)** CH_3-ZnCl

Hint : "more electronegativity difference, more the reactivity'.

A2

Strong nucleophile

(A) CH_3K

2.5 0.8

0.9

} Ionic bond

(B) CH_3Na

1

(C) CH_3Li

1.2

(D) CH_3MgBr

1.9

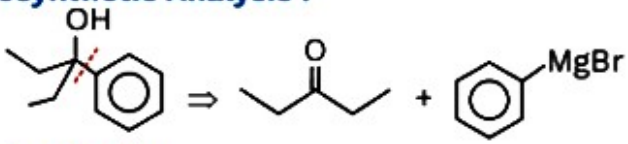
(E) CH_3-CuCl

2.5 2.1

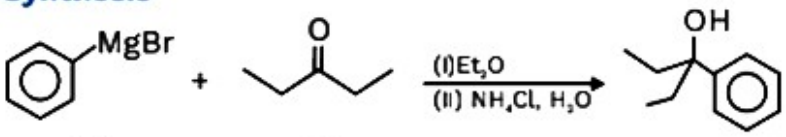
(F) CH_3-ZnCl

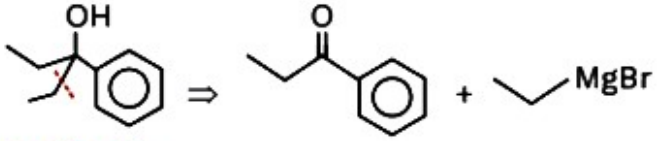
} Weak nucleophile

Retrosynthetic Analysis :

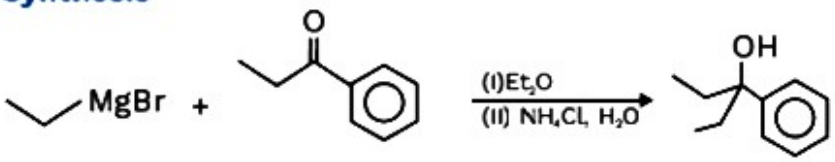
1. 

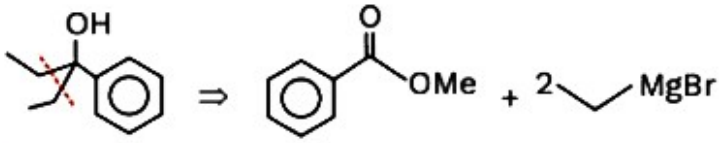
Synthesis



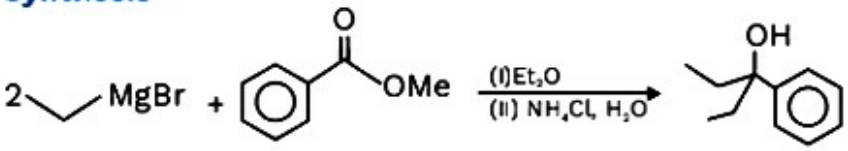
2. 

Synthesis



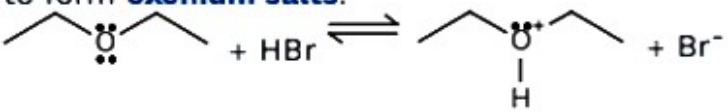
3. 

Synthesis



Cleavage of Ether

The oxygen of the ether linkage makes ethers weakly basic. Ethers can react with proton donors to form **oxonium salts**:



An oxonium salt

The first step in the cleavage of an ether by HI or HBr is protonation of the ether oxygen. This converts the very basic RO- leaving group into the less basic ROH leaving group.

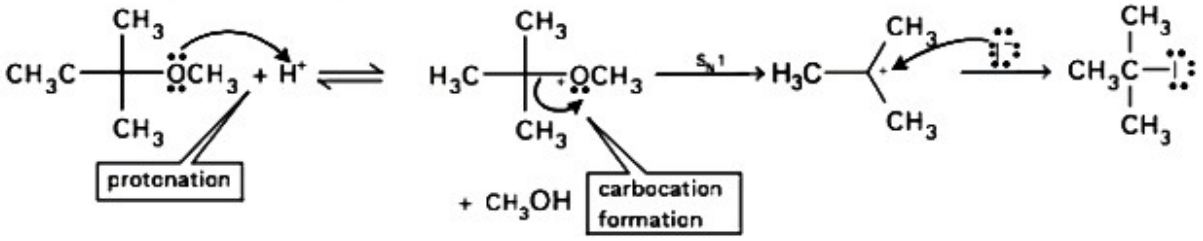
What happens next in the mechanism depends on the structure of the ether

Tips and Trick

Heating dialkylethers with very strong acids (HI, HBr, and H_2SO_4) causes them to undergo reactions in which the carbon–oxygen bond breaks.

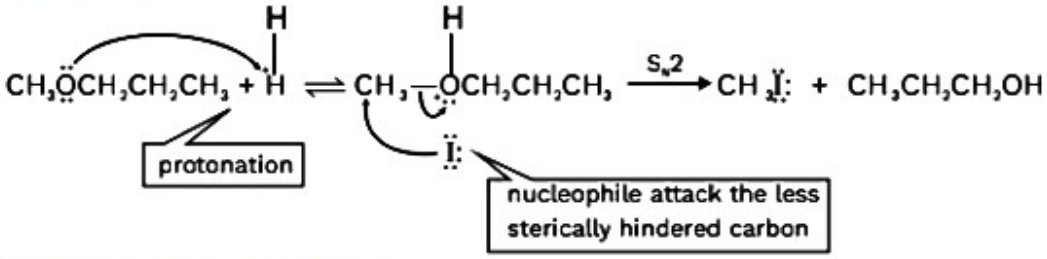
Concept for S_N1

If departure of the leaving group creates a relatively stable carbocation (e.g., a tertiary carbocation), an SN_1 reaction occurs—the leaving group departs, and the halide ion combines with the carbocation.



Concept for s_n2

If departure of the leaving group would create an unstable carbocation (e.g., a methyl, vinyl, aryl, or primary carbocation), the leaving group cannot depart. It has to be displaced by the halide ion. In other words, SN_2 an reaction occurs.

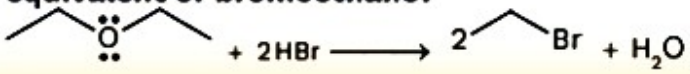


Points to Remember

- The cleavage of ethers by HI or HBr occurs more rapidly if the reaction can take place by an pathway.
- If the instability of the carbocation requires the reaction to follow an SN_2 pathway, cleavage will be more rapid with HI than with HBr because I is a better nucleophile than Br^-.
- Only a substitution product is obtained because the bases present in the reaction mixture (halide ions and) are too weak to abstract a proton in an E2 reaction, and any alkene formed in an E1 reaction would undergo electrophilic addition with HBr or HI to form the same alkyl halide as would be obtained from the substitution reaction.
- Concentrated HCl cannot cleave ethers because is too poor a nucleophile.

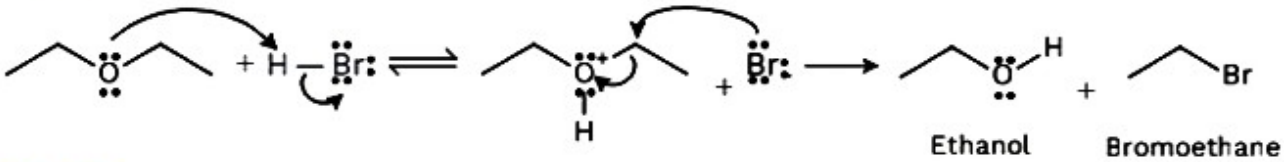
Questions

Q34 **How Excess HBr reacts with the ethanol produced to form the second molar equivalent of bromoethane?**

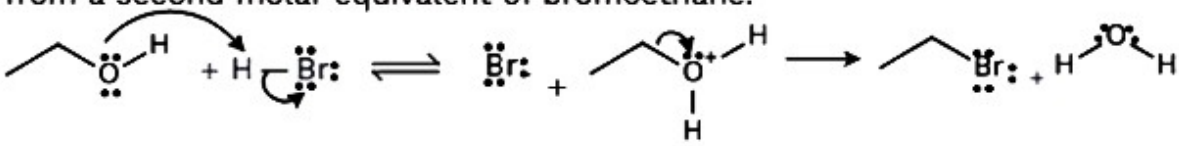


A34 A Mechanism for the Reaction

Step 1 :

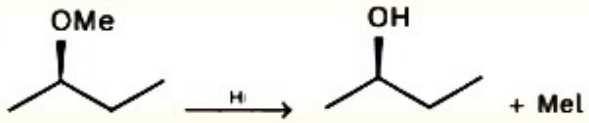


Step 2 : In step 2 the ethanol (just formed) reacts with HBr (present in excess) to from a second molar equivalent of bromoethane.

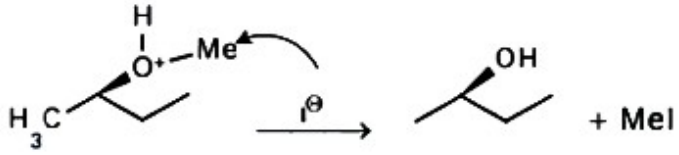


When mixed ethers are used, the alcohol and alkyl iodide that from depend on the nature of the alkyl groups.
Use mechanisms to explain the following observations:

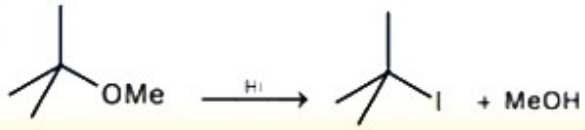
Q35



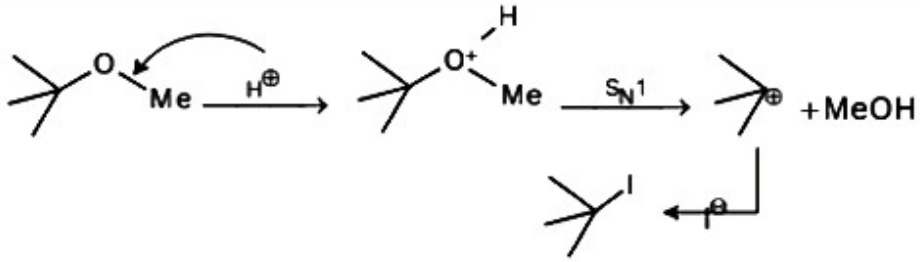
A35



Q36



A36



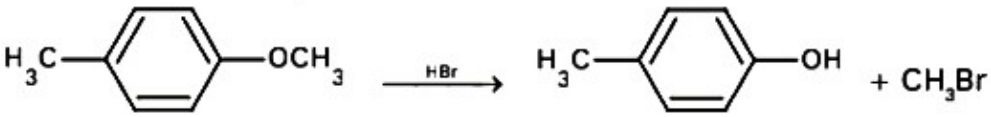
Cleavage of Alkyl Aryl Ethers

Phenol does not react to produce an aryl halide because the phenol carbon–oxygen bond is very strong and because phenyl cations do not form readily.

In other words, neither an SN2 nor an SN1 reaction can occur at the *sp*2 carbon of the phenol, but they can occur at the alkyl carbon of an ether.

Introduction

When alkyl aryl ethers react with strong acids such as HI and HBr, the reaction produces An alkyl halide and a phenol.



p – Methylanisole — 4 – Methylphenol — Bromomethane

Give the major products that would be obtained from heating each of the following ethers with HI :

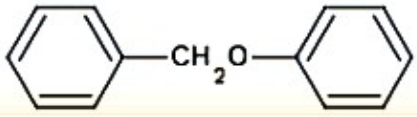
Q37 $CH_3CH = CHOCH_2CH_3$

A37

$$CH_3 - CH = CH - O - CH_2 - CH_3 \xrightarrow{H^{\oplus}} H_3C-CH=CH-\overset{H}{\underset{Et}{O^{+}}}$$

$$\xrightarrow{I^{\ominus}} CH_3-CH=CH-OH + Et-I$$

Q38

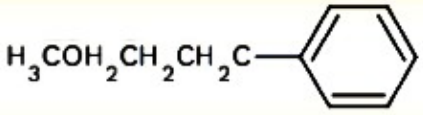


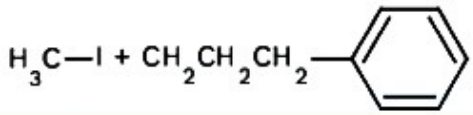
A38

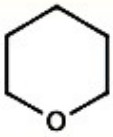
$$Ph - CH_2 - O - Ph \xrightarrow{H^{\oplus}} Ph - CH_2 - \overset{H}{\underset{\oplus}{O}} - Ph$$

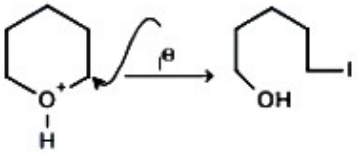
$$\longrightarrow Ph - \overset{\oplus}{C}H_2 + Ph - OH \xrightarrow{I^{\ominus}} Ph - CH_2 - I + Ph - OH$$

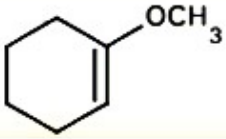
Q39

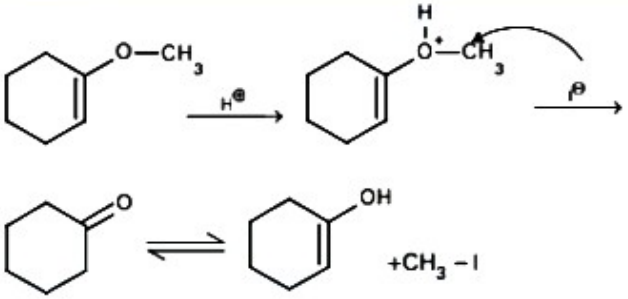


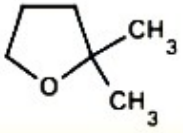
A39 

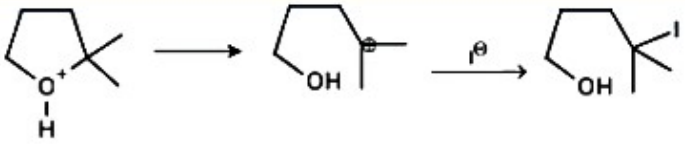
Q40 

A40 

Q41 

A41 

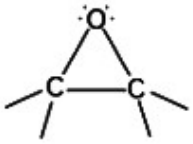
Q42 

Q42 

Epoxides

Epoxides are cyclic ethers with three-membered rings. In IUPAC nomenclature epoxides are called **oxiranes**.

The simplest epoxide has the common name ethylene oxide :



An epoxide



IUPAC name : oxirane
Common name : ethylene oxide

Tips and Trick

Ether is bivalent Functional group that's why higher alkyl group is deciding.

(A) 1° S_N2

(B) 3° S_N1

(C) 1° + Reso stable cation S_N1

(D) 2° → Reagent Dependent → Conc. S_N2 / dil. S_N1

(E) Ph ¾→ No Reaction

Stereochemistry of Epoxidation

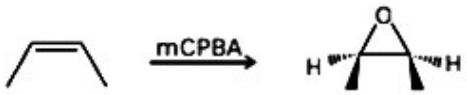
Although an epoxide and an ether have the same leaving group, epoxides are much more reactive than ethers in nucleophilic substitution reactions because the strain in the three-membered ring is relieved when the ring opens.

Example:

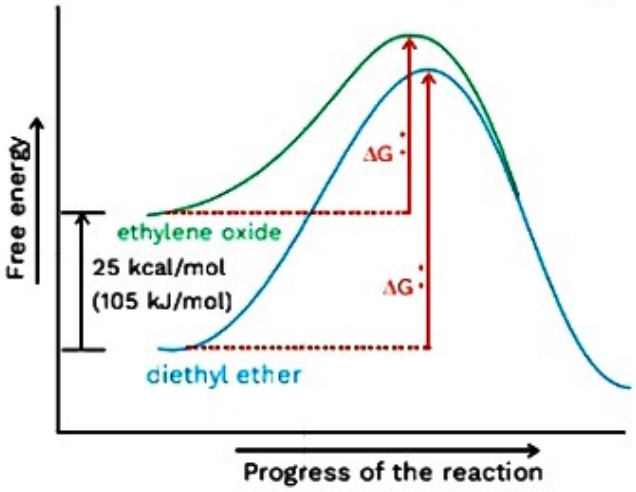


trans-2-Butene Enantiomeric trans-2,3-dimethyloxiranes

Example:

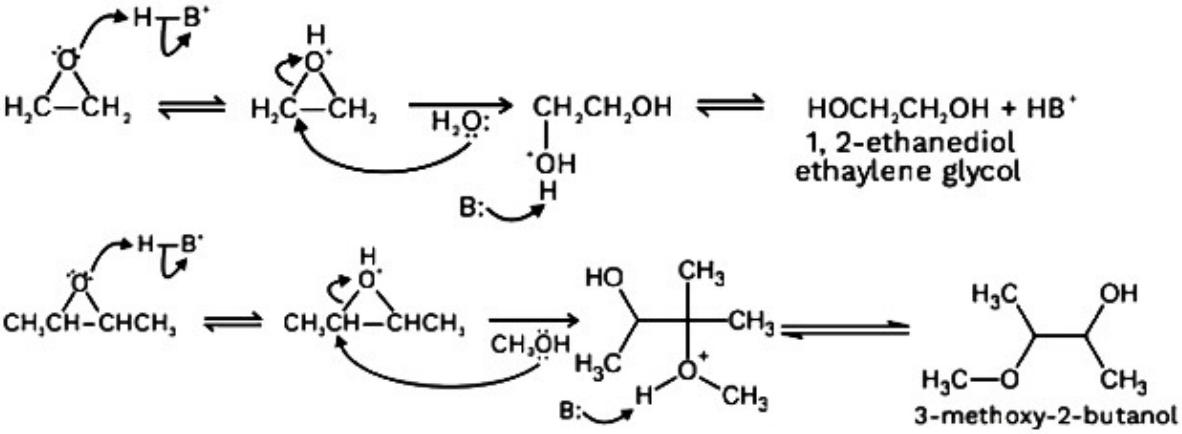


cis-2-Butene cis-2,3-Dimethyloxirane (a meso compund)



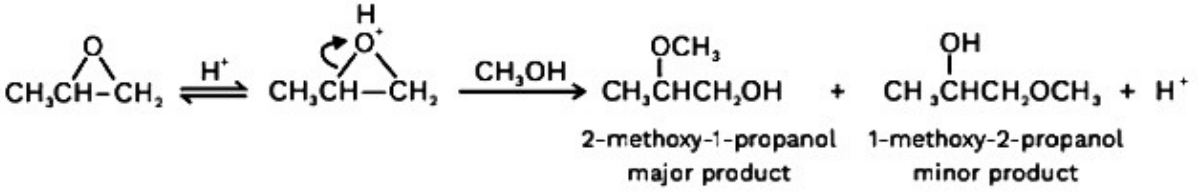
Points to remember

Protonated epoxides are so reactive that they can be opened by poor nucleophiles, such as H_2O and alcohols. (HB^+ is any acid in the reaction mixture; B: is any base.

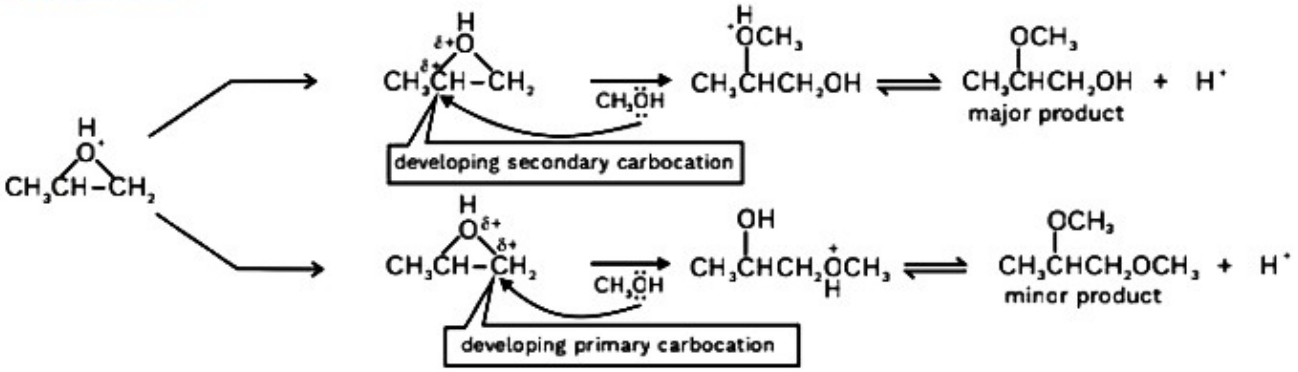


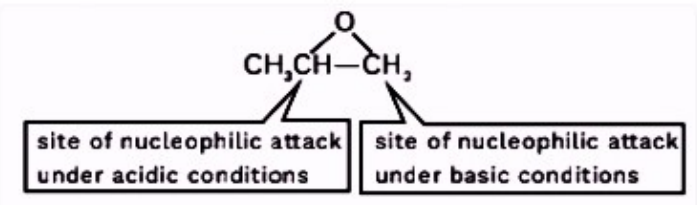
Note :

If different substituents are attached to the two carbons of the protonated epoxide (and the nucleophile is something other than H_2O), the product obtained from nucleophilic attack on the 2-position of the oxirane ring will be different from that obtained from nucleophilic attack on the 3-position. The major product is the one resulting from nucleophilic attack on the *more substituted* carbon



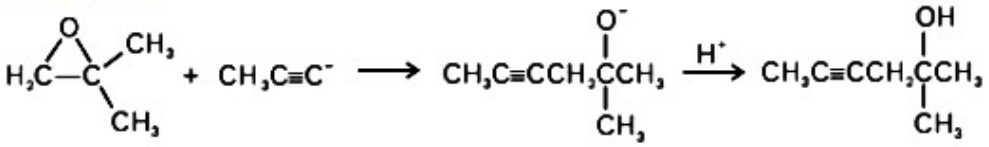
Mechanism :

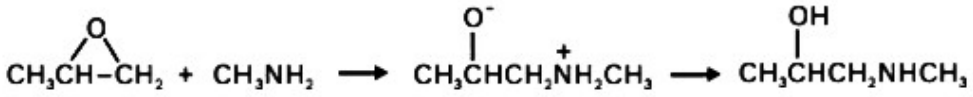




After the nucleophile has attacked the epoxide, the alkoxide ion can pick up a proton from the solvent or from an acid added after the reaction is over.
Epoxides are synthetically useful reagents because they can react with a wide variety of nucleophiles, leading to the formation of a wide variety of products.

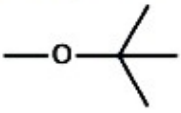
Example :





Question
Obtain desired product of following ether cleavage reaction

Q43 **Compound**



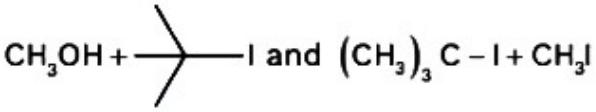
Case-1

Structure of product (when react with conc. HI)

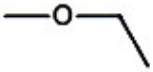
Case-2

Structure of product (when react with conc. HI excess)

A43



Q44. **Compound**



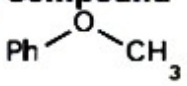
Case-1

Structure of product react with conc. HI)

Case-2

(when Structure of product (when react with conc. HI excess)

A44 $EtOH + CH_3I$ and $CH_3I + PhOH$

Q45. **Compound** 

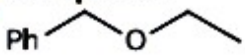
Case-1

Structure of product (when react with conc. HI)

Case-2

Structure of product (when react with conc. HI excess)

A45 $CH_3I + PhOH$ and $CH_3I + PhOH$

Q46. **Compound** 

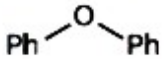
Case-1

Structure of product (when react with conc. HI)

Case-2

Structure of product (when react with conc. HI excess)

A46 $PhCH_2I + EtOH$ and $PhCH_2I$ + Et-I

Q47 **Compound** 

Case-1

Structure of product (when react with conc. HI)

Case-2

Structure of product (when react with conc. HI excess)

A47 No reaction

Q48. **Compound**
PhOH

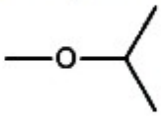
Case-1

Structure of product (when react with conc. HI)

Case-2

Structure of product (when react with conc. HI excess)

A48 No reaction

Q49 **Compound** 

Case-1

Structure of product (when react with conc. HI)

Case-2

Structure of product (when react with conc. HI excess)

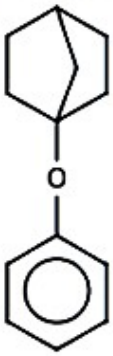
A49 $CH_3I + CH_3CH(OH)CH_3$ and $CH_3I + CH_3CH(I)CH_3 + H_2O$

Q50

Compound	Case-1	Case-2
$CH_3 - CH_2 - OH$	**Structure of product (when react with conc. HI)**	**Structure of product (when react with conc. HI excess)**

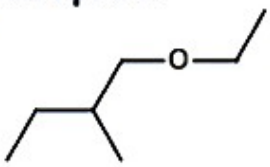
A50 $CH_3CHI + H_2O$ and CH_2CH_2I

Q51

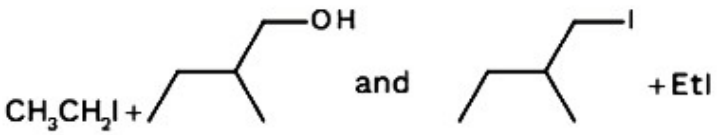
Compound	Case-1	Case-2
	**Structure of product (when react with conc. HI)**	**Structure of product (when react with conc. HI excess)**

A51 No reaction

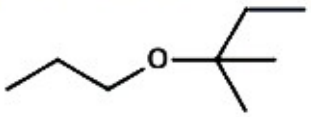
Q52

Compound	Case-1	Case-2
	**Structure of product (when react with conc. HI)**	**Structure of product (when react with conc. HI excess)**

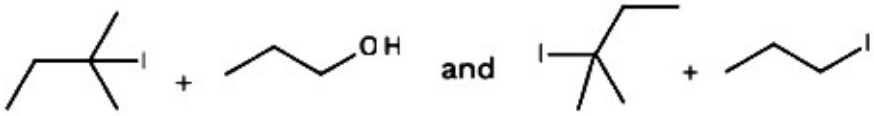
A52



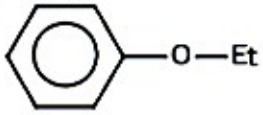
Q53 **Compound** **Case-1** **Case-2**



Structure of product (when react with conc. HI) **Structure of product (when react with conc. HI excess)**

A53 

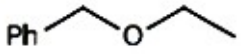
Q54 **Compound** **Case-1** **Case-2**



Structure of product (when react with conc. HI) **Structure of product (when react with conc. HI excess)**

A54 EtI+PhOH and EtI+PhOH

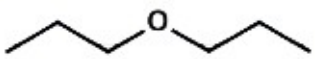
Q55 **Compound** **Case-1** **Case-2**



Structure of product (when react with conc. HI) **Structure of product (when react with conc. HI excess)**

A55 

Q56 **Compound** **Case-1** **Case-2**



Structure of product (when react with conc. HI) **Structure of product (when react with conc. HI excess)**

A56 

General methods of preparation of Ether

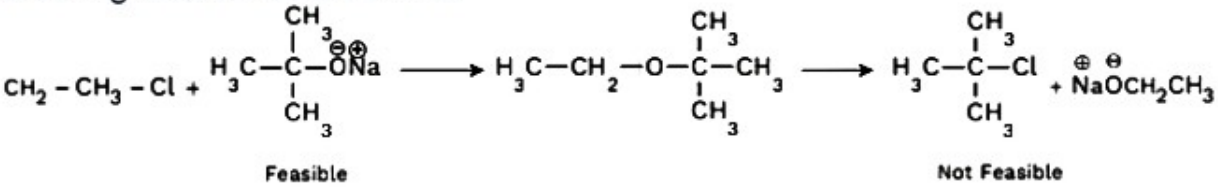
(a) Dehydration of Alcohols : The ethers are prepared by reactions of the corresponding alcohols with sulfuric Acid. Since a molecule of water is lost for every pair of molecules, the reaction is kind of dehydration.

$$2ROH \xrightarrow{H_2SO_4,\ Heat} R-O-R + H_2O$$

(b) Williamson Synthesis : In this reaction sodium or potassium alkoxide is heated with an alkyl halide. The mechanism is S_N2

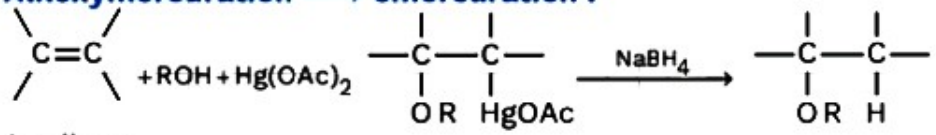
$$R\overset{\ominus}{O}Na + R'-X \longrightarrow R\overset{\delta-}{O}- - \overset{\delta+}{R} - - \overset{\delta-}{X} \longrightarrow R-O-R' + \overset{+}{N}a\ \overset{-}{X}$$

This method is particularly useful for preparing mixed ether, and it is best to use the alkoxide of the secondary or tertiary alcohol, and primary alkyl halide because secondary and tertiary alkyl halide readily undergo E2 elimination in the presence of a strong base to form alkenes.



Feasible — Not Feasible

(c) Alkoxymercuration ⟶ emercuration :



An alkene

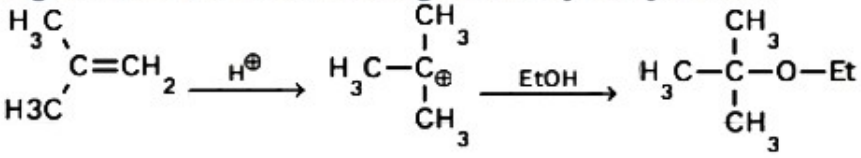
(d) Reaction of alkyl halides with any silver oxide :

Ether can also be prepared by heating an alkyl halide with dry silver oxide

$2CH_3–CH_2–Br + Ag_2O \rightarrow CH_3CH_2–O–CH_2CH_3 + 2\ AgBr$

(e) Ethers may be prepared by the addition of alcohols to alkenes in the presence of acid.

e.g, ethanol and isobutene give t-butyl ethyl ether.



Reactions of Ethers :

1. Ethers dissolve in concentrated solutions of strong inorganic acids to form oxonium salts, i.e. ethers behave as Bronsted Lowry bases.

$$R_2O + H_2SO_4 \rightleftharpoons [R_2OH]^+ HSO_4^-$$

2. When heated with dilute sulphuric acid under pressure, ethers form the corresponding alcohols.

$$R_2O + H_2O \xrightarrow{H_2SO_4} 2ROH$$

Tips and Trick

This reaction gives very good yields with alkenes, which can product-carborium ions and is very useful for preparing mixed ethers.

3. **Cleavage by Acids**

Ethers are comparatively unreactive compounds. The ether linkage is quite stable toward bases, oxidizing agents, and reducing agents. But ethers undergo cleavage by acids.

$$\mathbf{R{-}O{-}R' + HX \rightarrow R{-}X + R' {-}OH \xrightarrow{HX} R'{-}X}$$

Reactivity of HX : HI > HBr > HCl

Cleavage takes place only under quite vigorous conditions:

Concentrated acids (usually HI or HBr) and high temperatures. The initial reaction between an ether and an acid is undoubtedly formation of the protonated ether. Cleavage involves nucleophilic attack by halide ion on this protonated ether, with displacement of the weakly basic alcohol molecule :

$$R-\ddot{O}-R' + HX \rightleftharpoons \underset{H}{R-\overset{+}{O}-R} + X \xrightarrow[\text{Slow}]{S_N2} R-OH + R'X$$

$$\downarrow S_N1, \text{ slow}$$

$$R-OH + {}'R^+ \xrightarrow{X^\ominus} R-OH + R'X$$

Such a reaction occurs much more readily than displacement of the strongly basic alkoxide ion from the neutral ether.

$$R-O-R' + X \not\rightarrow R-X + R'O-$$

Strong base: poor leaving group

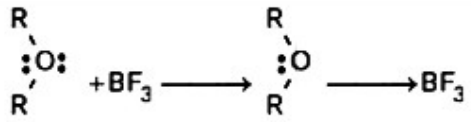
The above reaction will follow S_N1 or S_N2 depending upon conditions and the structure of the ether. As one might expect a primary alkyl group tends to undergo S_N2 displacement, where as a tertiary alkyl group tends to undergo S_N1 displacement.

Points to remember

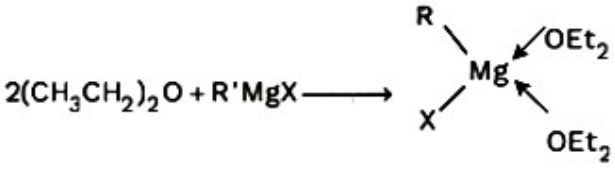
The above reaction will follow S_N1 or S_N2 depending upon conditions and the structure of the ether. As one might expect a primary alkyl group tends to undergo S_N2 displacement, where as a tertiary alkyl group tends to undergo S_N1 displacement.

Formation of Co-ordination complexes :

Being Lewis bases, ethers form co-ordinate complexes known as etherates with Lewis acids such as BF_3, $AlCl_3$, $FeCl_3$, Grignard reagents etc.



Boron triflouride etherate



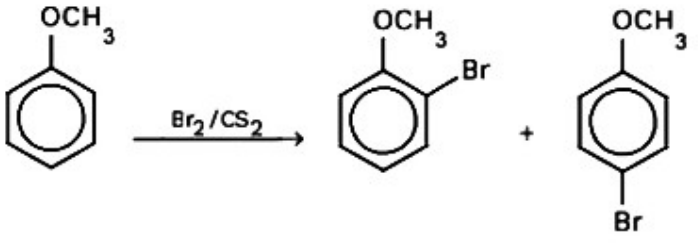
Grignard reagent etherate

Due to the formation of these complexes, Grignard reagents dissolve in ether. That is why Grignard reagents are usually prepared in ethers. However they cannot be prepared in other organic solvents like benzene, because benzene has no lone pairs of electrons and hence cannot form complexes with them.

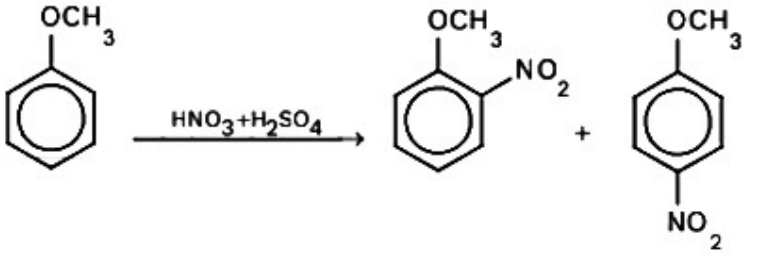
4. Electrophilic substitution Reactions :

Like phenols, aromatic ethers also undergo electrophilic substitution reactions in the benzene ring at ortho and para positions.

For example,



Anisole o–Bromoanisole p–Bromoanisole



Anisole o–nitro anisole p–nitro anisole